

Correlation And Covariance

Q1. Define Covariance and explain how it differs from Correlation in terms of scale and interpretation

Ans. It measures whether two variables move together or opposite to each other.

- Both go up together → Positive covariance
- One goes up, other goes down → Negative covariance
- No pattern at all → Near Zero

The number it gives you is meaningless without context. If scores are out of 100, covariance might be 24. If scores are out of 1000, covariance becomes 240. The relationship didn't change — but the number did. So you can never really compare or interpret it directly.

Correlation — The Fix

Correlation takes covariance and standardizes it by dividing by both variables' standard deviations. This removes the effect of scale and squishes everything into -1 to $+1$.

- $+1$ → Perfect positive relationship
- -1 → Perfect negative relationship
- 0 → No relationship
- $+0.85$ → Strong positive (you can actually say this with confidence)

Q2. What does a positive, negative, and zero covariance indicate about the relationship between two variables?

Ans. Positive Covariance

When one variable increases, the other also increases. Study hours & exam scores Students who study more → score higher. Both moving in the same direction.

Negative Covariance

When one variable increases, the other decreases. Stress levels & sleep hours More stress → less sleep. They move in opposite directions.

Zero Covariance →→

No consistent relationship between the two variables. One changes but the other does nothing predictable. Shoe size & intelligence, Your shoe size has absolutely nothing to do with how smart you are. No pattern, no relationship.

Q3. Discuss the limitations of covariance as a measure of relationship between two variables. Why is correlation preferred in many cases?

Ans. Limitation 1 — Scale Dependent

Covariance value changes when you change the units of measurement, even if the actual relationship stays the same. Height & Weight relationship

- Measure height in cm → covariance = 150

Limitation 2 – No Fixed range

Covariance can be any number from $-\infty$ to $+\infty$. There's no standard boundary, so you can never say if a value is big or small without extra context.

Limitation 3 — Can't Compare Across Different Datasets

Since covariance depends on scale, you cannot compare covariances from two different studies or datasets meaningfully.

Example:

- Covariance between income & spending = 10,000
- Covariance between age & health score = 18

Limitation 4 — Only Measures Linear Relationship

Covariance only detects straight-line (linear) relationships. If two variables have a curved or complex relationship, covariance might show zero — making you think there's no relationship when there actually is one.

Limitation 5 — Hard to Interpret Directly

A non-statistician looking at a covariance of -347.6 has absolutely no idea what that means. It's not intuitive or self-explanatory.

Covariance tells you if a relationship exists and in which direction. But correlation tells you how strong, which direction, and in a way that is universally understandable and comparable — which is why it's almost always preferred in real analysis.

Q4. Explain the difference between Pearson's correlation coefficient and Spearman's rank correlation coefficient. When would you prefer to use Spearman's correlation?

Ans. Pearson vs Spearman Correlation

Pearson's Correlation Coefficient

It measures the linear relationship between two variables — assuming both variables are normally distributed and the data is continuous.

It works on actual values of the data.

Example: Height & Weight

As height increases, weight increases in a straight-line pattern. Pearson works perfectly here.

- Range: -1 to $+1$
- Works best when data is clean, normal, and linear

Spearman's Rank Correlation

Instead of working on actual values, it converts data into ranks first and then measures the relationship between those ranks.

Example: Class ranking & Performance rating

Student ranked 1st tends to get highest performance rating. Spearman compares these ranks, not raw numbers.

- Range: -1 to $+1$ (same as Pearson)
- Works on ranked or ordinal data
- Captures any monotonic relationship (not just linear)

When to prefer spearman's correlation

1. Ordinal data

When your data is ranked or ordered but not precisely measured

Examples: Customer satisfaction rating (1-5 stars) Movie rankings, survey response

1. Ordinal Data

When your data is **ranked or ordered** but not precisely measured.

Example: Customer satisfaction ratings (1-5 stars), movie rankings, survey responses like "agree/disagree"
Like "Agree/Disagree"

2. Outliers present

Pearson gets heavily distorted by outliers spearman is based on ranks so one extreme value

Won't ruin the result

Example: income data where one billionaire is in the dataset – spearman handles this better

3. Non – normal distribution

When your data is skewed or not normally distributed, pearson's assumption is violated,
Spearman makes no such assumption.

4. Non – Linear but monotonic relationship

If the relationship is curved but still consistently going the one direction, spearman

Catches it, pearson might miss it.

Example: As experience increase, salary increases – but not at a constant rate. Spearman
Handles this better.

Q5. If the correlation coefficient between two variables X and Y is 0.85, interpret this value in context.

Can you infer causation from this value? Why or why not?

Ans. A correlation of +0.85 means there is a strong positive relationship between X and Y.

When X increases, Y also tends to increase — and this happens in a very consistent pattern.

It's close to $+1$ (perfect correlation) but not exactly 1, meaning the relationship is strong but not perfect — there
Are some variations.

Correlation Value	Interpretation
0.90 – 1.00	Very strong
0.85	Strong
0.50 – 0.70	Moderate
0.30 – 0.50	Weak
0.00 – 0.30	Very weak / No relationship

A correlation of 0.85 tells you the relationship is strong and positive — X and Y move together consistently. But it tells you nothing about why. To claim causation, you need proper experiments and deeper analysis, not just a correlation number.

Q6. Using the dataset below, calculate the covariance between X and Y

X	2	4	6	8
Y	3	7	5	10

Ans. Calculating Covariance Between X and Y

Step 1 — Find the Means

Mean of X:

$$\bar{X} = \frac{2 + 4 + 6 + 8}{4} = \frac{20}{4} = 5$$

Mean of Y:

$$\bar{Y} = \frac{3 + 7 + 5 + 10}{4} = \frac{25}{4} = 6.25$$

Step 2 — Apply Covariance Formula

$$\begin{aligned} Cov(X, Y) &= \frac{\sum(X - \bar{X})(Y - \bar{Y})}{n} \\ Cov(X, Y) &= \frac{9.75 + (-0.75) + (-1.25) + 11.25}{4} \\ Cov(X, Y) &= \frac{19}{4} \\ \boxed{Cov(X, Y) = 4.75} \end{aligned}$$

The covariance is +4.75, which is positive — meaning X and Y tend to move in the same direction. When X increases, Y also tends to increase. However, since covariance has no fixed scale, we can only confirm the direction, not the strength of this relationship.

Q7. Compute the Pearson correlation coefficient between variables A and B:

A	10	20	30	40	50
B	8	14	18	24	28

Ans. Pearson Correlation Coefficient Between A and B

Step 1 — Find the Means

Mean of A:

$$\bar{A} = \frac{10 + 20 + 30 + 40 + 50}{5} = \frac{150}{5} = 30$$

Mean of B:

$$\bar{B} = \frac{8 + 14 + 18 + 24 + 28}{5} = \frac{92}{5} = 18.4$$

Step 2 — Apply Pearson Formula

$$\begin{aligned} r &= \frac{\sum(A - \bar{A})(B - \bar{B})}{\sqrt{\sum(A - \bar{A})^2 \times \sum(B - \bar{B})^2}} \\ r &= \frac{500}{\sqrt{1000 \times 251.20}} \\ r &= \frac{500}{\sqrt{251200}} \\ r &= \frac{500}{501.19} \\ \boxed{r = 0.9976 \approx 0.998} \end{aligned}$$

The Pearson correlation of 0.998 tells us that A and B have an almost perfect positive linear relationship. As A increases, B increases in a very consistent and predictable pattern. This is an extremely strong correlation — nearly a straight line if plotted on a graph.

Q8. The following table shows heights (in cm) and weights (in kg) of 5 students. Find the correlation Coefficient between height and weight.

Height	150	160	165	170	180
Weight	50	55	58	62	70

Ans. Pearson Correlation Between Height and Weight

Step 1 — Find the Means

Mean of Height:

$$\bar{H} = \frac{150 + 160 + 165 + 170 + 180}{5} = \frac{825}{5} = 165$$

Mean of Weight:

$$\bar{W} = \frac{50 + 55 + 58 + 62 + 70}{5} = \frac{295}{5} = 59$$

Step 2 — Apply Pearson Formula

$$r = \frac{\sum(H - \bar{H})(W - \bar{W})}{\sqrt{\sum(H - \bar{H})^2 \times \sum(W - \bar{W})^2}}$$

$$r = \frac{335}{\sqrt{500 \times 228}}$$

$$r = \frac{335}{\sqrt{114000}}$$

$$r = \frac{335}{337.63}$$

$$r = 0.9922 \approx 0.99$$

A correlation of 0.99 means taller students are almost perfectly consistently heavier too. If you plotted this on a graph, the points would fall almost exactly on a straight line. This is a very strong and reliable relationship between Height and Weight in this dataset.

Q9. Given the dataset below, determine whether there is a positive or negative correlation between X and Y. (No need for exact calculation, just reasoning.)

X	1	2	3	4	5
Y	15	12	9	7	3

Ans. Observe the Pattern

- X goes 1 → 2 → 3 → 4 → 5 → consistently increasing
- Y goes 15 → 12 → 9 → 7 → 3 → consistently decreasing

Every time X goes up by 1, Y goes down significantly.

They are moving in completely opposite directions — this is a textbook case of:

The decrease in Y is very consistent and smooth — no random jumps or breaks in pattern. This suggests the negative correlation is not just negative but likely very strong, close to −1.

As X increases, Y consistently and steadily decreases. This is clearly a strong negative correlation — no calculation needed, the pattern speaks for itself.

Q10. Two investment portfolios have the following returns (%) over 5 years. Compute the covariance and correlation coefficient, and interpret whether the portfolios move together.

Year	Portfolio A	Portfolio B
1	8	6
2	10	9
3	12	11
4	9	8
5	11	10

Ans. Covariance & Correlation Between Portfolio A and B

Step 1 — Find the Means

Mean of A:

$$\bar{A} = \frac{8 + 10 + 12 + 9 + 11}{5} = \frac{50}{5} = 10$$

Mean of B:

$$\bar{B} = \frac{6 + 9 + 11 + 8 + 10}{5} = \frac{44}{5} = 8.8$$

Step 2 — Compute Covariance

$$Cov(A, B) = \frac{\sum(A - \bar{A})(B - \bar{B})}{n} = \frac{12}{5}$$

$Cov(A, B) = 2.4$

Step 3 — Compute Pearson Correlation

$$r = \frac{\sum(A - \bar{A})(B - \bar{B})}{\sqrt{\sum(A - \bar{A})^2 \times \sum(B - \bar{B})^2}}$$

$$r = \frac{12}{\sqrt{10 \times 14.80}}$$

$$r = \frac{12}{\sqrt{148}}$$

$$r = \frac{12}{12.166}$$

$r = 0.9864 \approx 0.99$

A correlation of 0.99 means Portfolio A and Portfolio B move almost perfectly together. When Portfolio A gives higher returns, Portfolio B almost always does too — and vice versa.

DRIVE LINK: https://drive.google.com/drive/folders/1ZlFjacMcn7A-Gi0Qmzbq18jmXQvYluqH?usp=drive_link